

7. ANALYTICAL TOOLS: APPLICATION AND FINDINGS

This section involves applying the right analytical tools, and analysing research findings and research data. The analytical tools used in all the research strands are presented and the main findings from each strand are documented.

7.1 DESCRIPTIVE STATISTICS SURVEY DATA

All six Likert dimensions were analyzed using descriptive statistical analysis for the quantitative survey (n=22). Measures calculated were: arithmetic mean, median, mode, standard deviation, minimum, maximum and percentage scores at 4–5 (favourable) and 1–2 (unfavourable). These are suitable for ordinal data (Likert) with a sample size of n=22, when parametric inferential statistics are not applicable.

Descriptive statistical approach proved to be especially useful in determining the shape of distribution. For example, Q6 (AI Adoption Support) has a mean of 2.91 which might superficially suggest moderate ambivalence. But the mode (1) and the complete distribution shows a bimodal distribution with equal peaks at scores 1, 2, and 4 much more analytically significant than the mean by itself. This is a clear indication that descriptive statistics should be used all the time, not just sometimes.

7.2 ALGORITHM BENCHMARKING ANALYSIS

A wide range of performance measures were used to test eight different variants of the algorithm: Precision, Recall, F1 Score, Accuracy, Inference Latency, and Novel Attack F1. The F1 Score was chosen as the primary decision metric as it provides a harmonic mean of the precision and recall which are both critical in the context of intrusion detection data sets which are often imbalanced, where false negatives (missed attacks) are as expensive as false positives (alert fatigue).

All supervised classifiers were trained after applying SMOTE (Chawla et al., 2002). If SMOTE is not used, the number of intrusion events in the minority class may be underrepresented, leading to artificially high accuracy by predicting the majority class. The supervised algorithm F1 scores improved by 0.12 to 0.19 after SMOTE, which confirmed the use of SMOTE as an important preprocessing step.

To minimize the risk of overfitting and to ensure that the performance measures reported are generalizable and not unique to the dataset, all supervised algorithms were evaluated using 5-fold cross validation. Latency benchmarking was added as a deployment-relevant metric, comparing all the results against the OT emergency shutdown time of 100-200ms.

7.3 CONCEPT DRIFT ANALYSIS

A 30-week simulated deployment of ICS was used for analysis of concept drift with 10 distinct drift events. The degradation rates were expressed as percentage loss from the pre-drift baseline F1 and the recovery times were expressed in days. The drift detection metric used was the

Population Stability Index (PSI) and the thresholds were as follows: $PSI > 0.2$ indicated a significant drift and required retraining of the model.

The most important finding was the degradation of the F1 by 29.4% after SCADA Protocol Configuration Change over 22 silent days. The word 'silent' is key; if this degradation were not automatically monitored by drift, it would not be noticed by operations personnel until a missed attack was detected.

7.4 ADVERSARIAL ATTACK ANALYSIS

The adversarial robustness was tested by running 10 attack types on the LSTM Autoencoder with 500 samples per attack type. A composite risk score was determined by applying weights to F1 degradation and evasion rate. The attacks were then ranked according to composite risk to offer an operational threat prioritization framework for RGS advisory clients.

7.5 TRIANGULATION ANALYSIS

Triangulation was used as an overarching analytical tool, bringing together results from the three research strands (survey, simulation, literature) to check for convergent validity. The triangulation matrix (Table 9) records the areas where all three strands intersect (where confidence increases), the areas where two strands intersect (moderate-high confidence), and the areas where there is a divergence (analytically significant finding in itself).

Research Finding	Survey Evidence	Simulation Evidence	Literature	Confidence
Current IDS inadequate for novel OT threats	Q4 mean 2.32/5; 68.2% score 1–2	Signature IDS novel $F1 \approx 0.0$	Stallings (2023); NCSC (2023)	Very High3/3 converge
AI outperforms Signature IDS	40.9% support AI (mean 2.91)	LSTM novel $F1 = 0.905$ vs Sig. $F1 \approx 0.0$	Gupta (2023); Darktrace (2024)	Very High3/3 converge
59.1% CI orgs experienced IT/OT incidents	Q5: 59.1% Yes	N/A	Ahmed (2023); NCSC (2023)	High2/3 converge

Research Finding	Survey Evidence	Simulation Evidence	Literature	Confidence
XAI broadly valued; Mode=5 (Essential)	Q9 mean 3.50; 54.5% score 4–5	N/A	Schneiderman (2020)	High2/3 converge
FL: awareness high; willingness low	Q11: 81.8% aware; Q12: 27.3% willing	FL Autoencoder F1=0.897 achievable	Kizza (2023); McMahan (2017)	High3/3 converge
Adversarial risk underestimated	Q13 mean 2.82; 45.5% score 1–2	PGD: F1 drops 0.905→0.335 (63%)	Mehrabi et al. (2022)	High3/3 converge
Concept drift: invisible vulnerability	Q15 open text confirms awareness	29.4% F1 drop; 22-day silent recovery	Gupta (2023) partial	Medium-High2.5/3